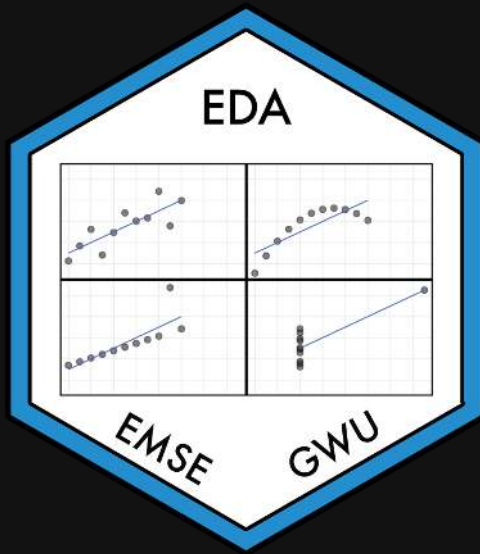




# Week 1: *Getting Started*

🏛️ EMSE 4572/6572: Exploratory Data Analysis

👤 John Paul Helveston

📅 August 27, 2025

# Week 1: *Getting Started*

1. Course Goal
2. Course Introduction
3. Break: Install Stuff
4. Quarto
5. Workflow & Reading In Data
6. Wrangling Data
7. Visualizing Data

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# Course 1: Intro to Programming for Analytics

## "Computational Literacy"

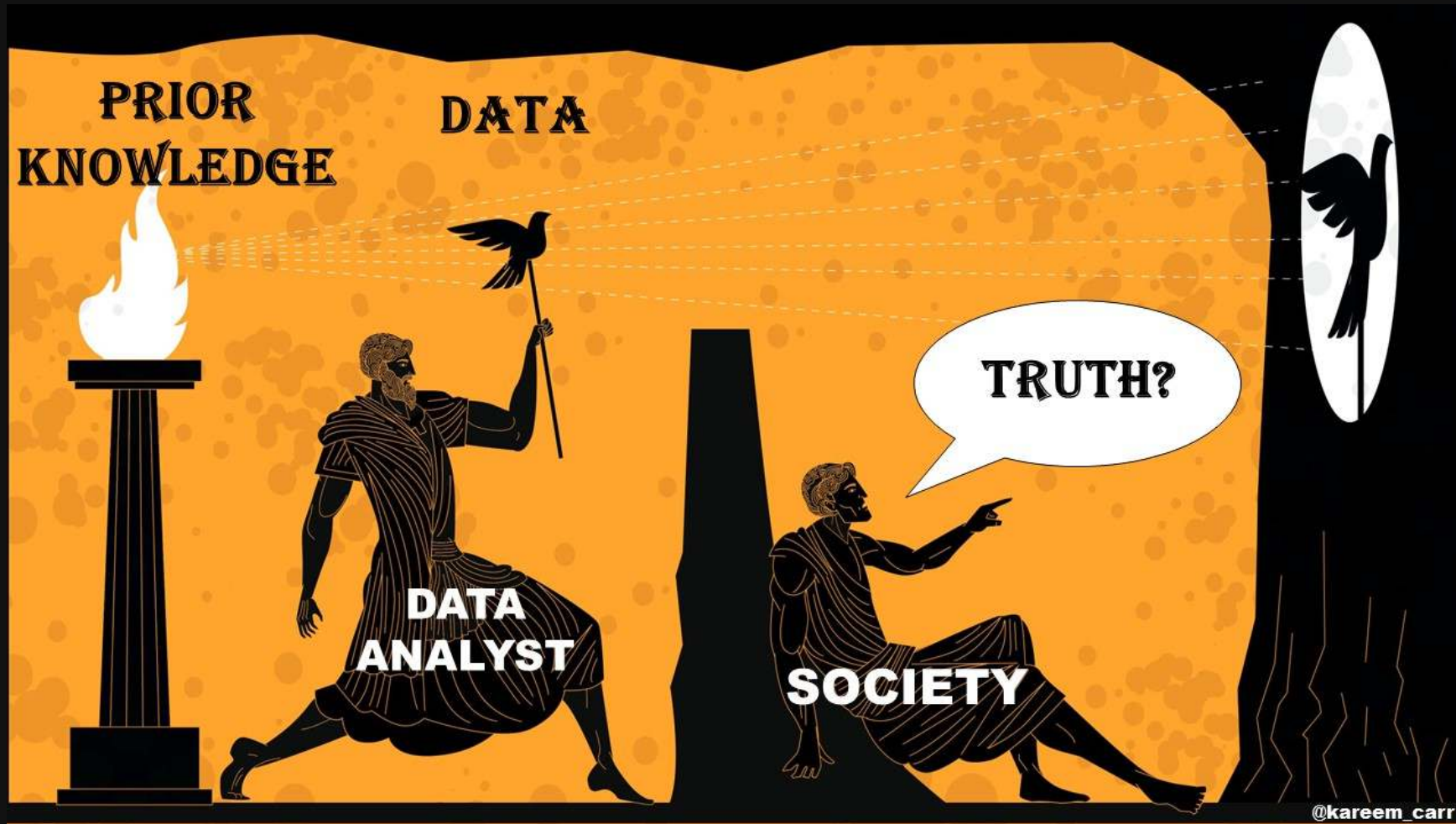
- Programming: Conditionals (if/else), loops, functions, testing, data types.
- Analytics: Data structures, import / export, basic data manipulation & visualization.

# Course 2: Exploratory Data Analysis

## "Data Literacy"

- Strategies for conducting an exploratory data analysis.
- Design principles for visualizing and communicating *information* extracted from data.
- Reproducibility: Reports that contain code, equations, visualizations, and narrative text.

# **Class goal:** translate *data* into *information*



# Class goal: translate *data* into *information*

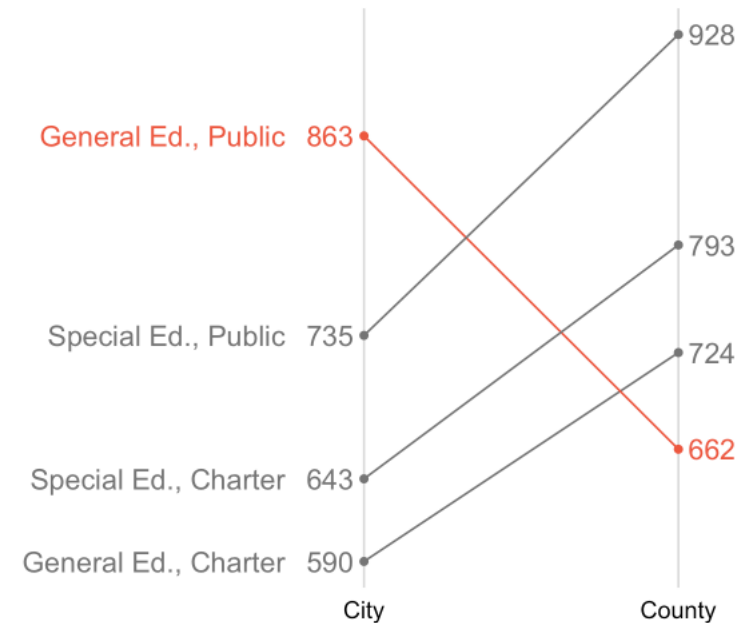
## Data

Average student engagement scores

| Class       | Type    | City | County |
|-------------|---------|------|--------|
| Special Ed. | Charter | 643  | 793    |
| Special Ed. | Public  | 735  | 928    |
| General Ed. | Charter | 590  | 724    |
| General Ed. | Public  | 863  | 662    |

## Information

Students in public, general education classes in county schools have surprisingly low engagement



# Data exploration: an iterative process

## Encode data:

```
engagement_data <- data.frame(  
  City    = c(643, 735, 590, 863),  
  County  = c(793, 928, 724, 662),  
  School  = c('Special Ed., Charter', 'Special Ed., Pu  
              'General Ed., Charter', 'General Ed., Pu  
engagement_data
```

```
#>   City County      School  
#> 1  643    793 Special Ed., Charter  
#> 2  735    928 Special Ed., Public  
#> 3  590    724 General Ed., Charter  
#> 4  863    662 General Ed., Public
```

## Re-format data for plotting:

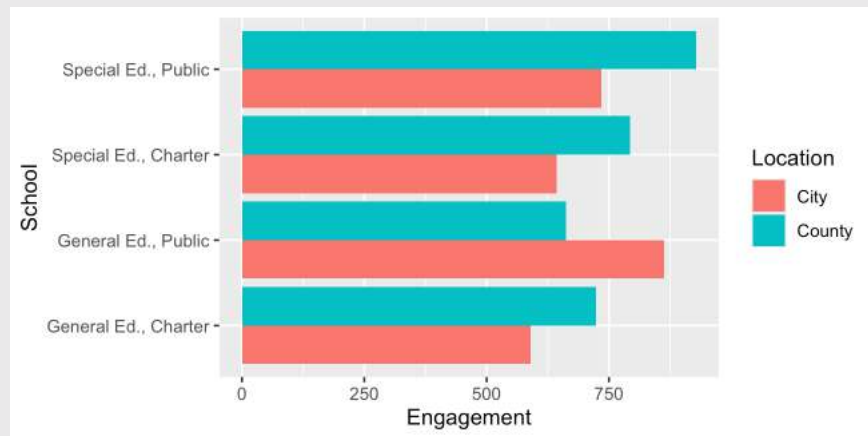
```
engagement_data <- engagement_data %>%  
  gather(Location, Engagement, City:County) %>%  
  mutate(Location = fct_relevel(  
    Location, c('City', 'County'))  
engagement_data
```

```
#>      School Location Engagement  
#> 1 Special Ed., Charter      City      643  
#> 2 Special Ed., Public      City      735  
#> 3 General Ed., Charter      City      590  
#> 4 General Ed., Public      City      863  
#> 5 Special Ed., Charter     County      793  
#> 6 Special Ed., Public     County      928  
#> 7 General Ed., Charter     County      724  
#> 8 General Ed., Public     County      662
```

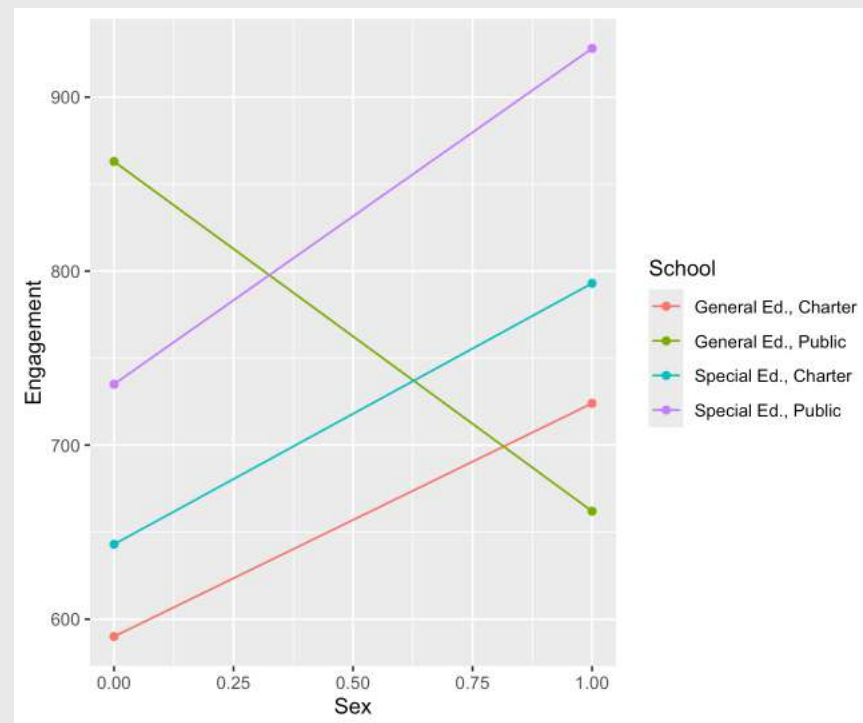
# Data exploration: an iterative process

Initial exploratory plotting:

```
engagement_data %>%  
  ggplot() +  
  geom_col(aes(x = Engagement, y = School,  
              fill = Location),  
          position = 'dodge')
```



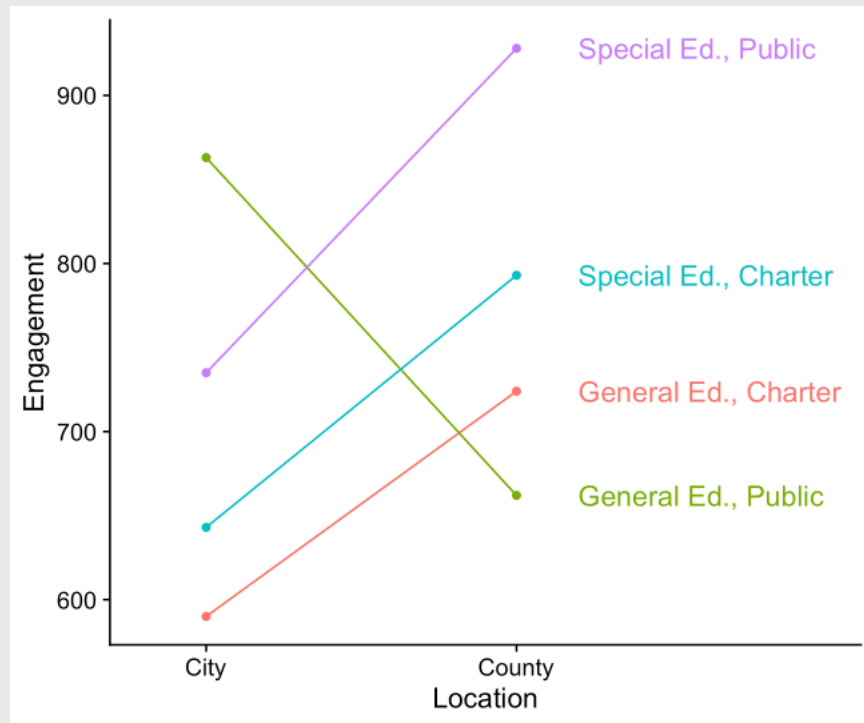
More exploratory plotting:  
highlight difference



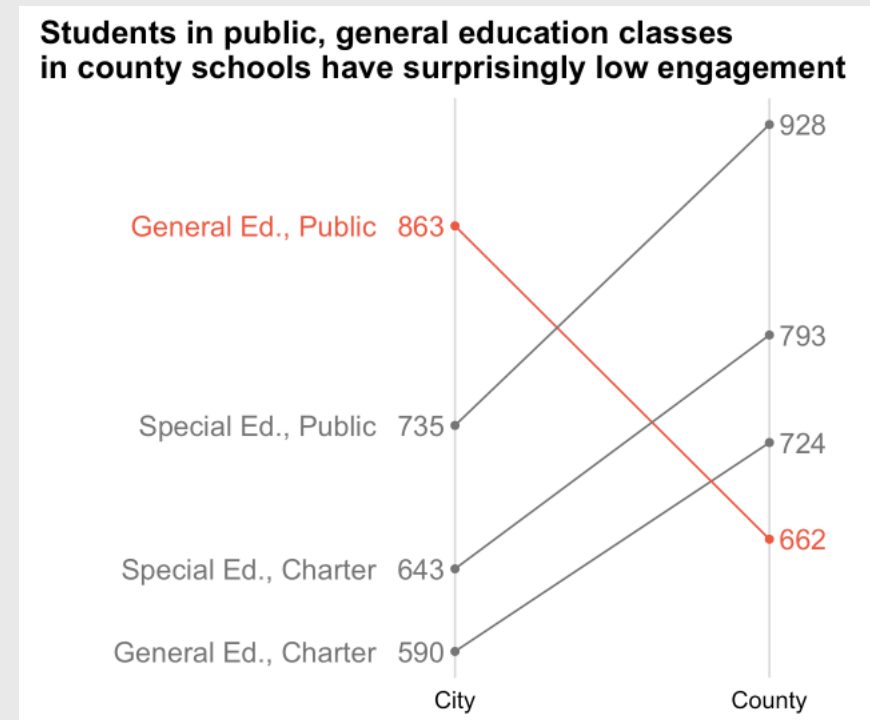


# Data exploration: an iterative process

Directly label figure:



Remove unnecessary axes, change colors, fix labels:



# A fully reproducible analysis

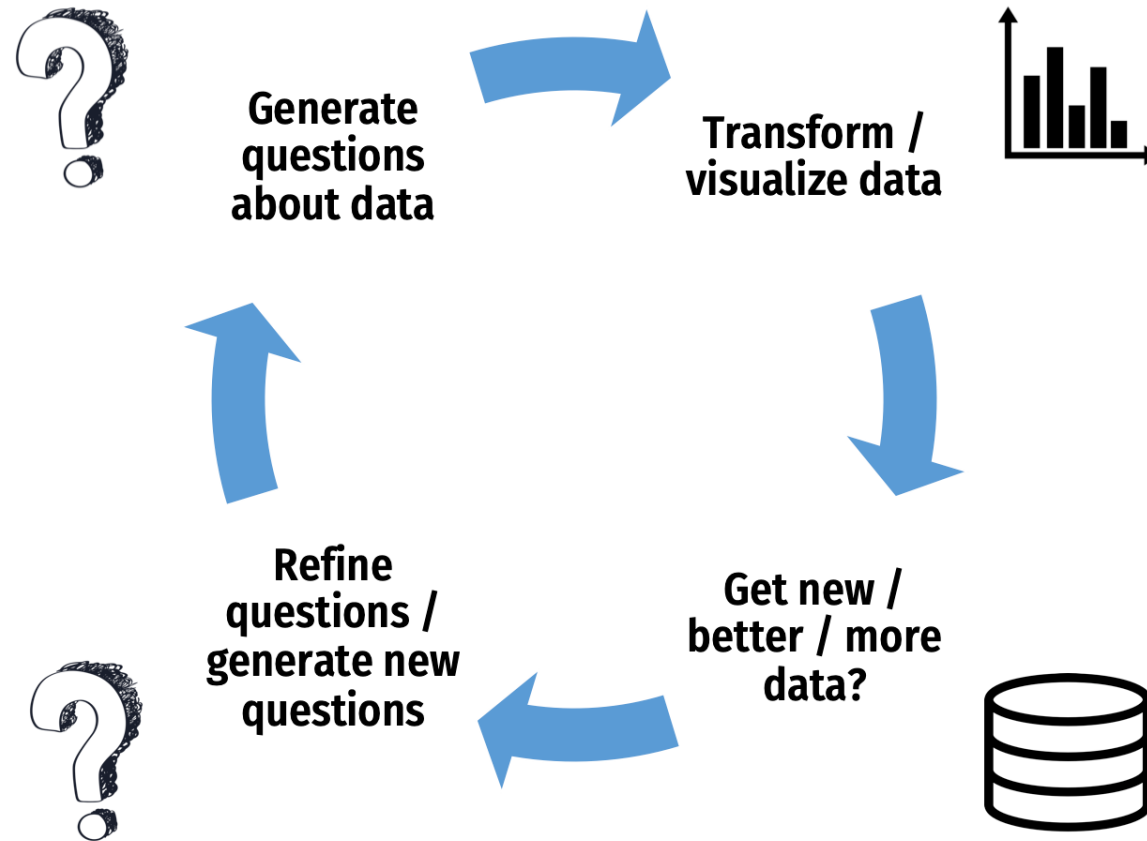
Code

Plot

```
data <- data.frame(
  City = c(643, 735, 590, 863),
  County = c(793, 928, 724, 662),
  School = c('Special Ed., Charter', 'Special Ed., Public',
             'General Ed., Charter', 'General Ed., Public'),
  Highlight = c(0, 0, 0, 1)) %>%
gather(Location, Engagement, City:County) %>%
mutate(
  Location = fct_relevel(Location, c('City', 'County')),
  Highlight = as.factor(Highlight),
  x = ifelse(Location == 'County', 1, 0))
```

```
plot <- ggplot(data, aes(x = x, y = Engagement, group = School, color = Highlight))
  geom_point() +
  geom_line() +
  scale_color_manual(values = c('#757575', '#ed573e')) +
  labs(x = 'Sex', y = 'Engagement',
       title = paste0('Students in public, general education classes\n',
                      'in county schools have surprisingly low engagement')) +
  scale_x_continuous(limits = c(-1.2, 1.2), labels = c('City', 'County'),
                    breaks = c(0, 1)) +
  geom_text_repel(aes(label = Engagement, color = as.factor(Highlight)),
                 data = subset(engagement, Location == 'County'),
                 size = 5,
                 nudge_x = 0.1,
                 segment.color = NA) +
  geom_text_repel(aes(label = Engagement, color = as.factor(Highlight)),
                 data = subset(engagement, Location == 'City'),
                 size = 5,
                 nudge_x = -0.1,
                 segment.color = NA) +
  geom_text_repel(aes(label = School, color = as.factor(Highlight)),
                 data = subset(engagement, Location == 'City'),
                 size = 5,
                 nudge_x = -0.25,
                 hjust = 1,
                 segment.color = NA) +
  theme_cowplot() +
  background_grid(major = 'x') +
  theme(axis.line = element_blank(),
        axis.title.x = element_blank(),
        axis.title.y = element_blank(),
        axis.text.y = element_blank(),
        axis.ticks = element_blank(),
        legend.position = 'none')
```

# Data exploration: an iterative process



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# Meet your instructor!



John Helveston, Ph.D.

- 2025 - Present: Associate Professor, EMSE
- 2018 - 2025: Assistant Professor, EMSE
- 2016-2018: Postdoc at [Institute for Sustainable Energy](#), Boston University
- 2016: PhD in Engineering & Public Policy at Carnegie Mellon University
- 2015: MS in Engineering & Public Policy at Carnegie Mellon University
- 2010: BS in Engineering Science & Mechanics at Virginia Tech
- Website: [www.jhelvy.com](http://www.jhelvy.com)

# Meet your tutors!



## **Pingfan Hu**

- Graduate Teaching Assistant (GTA)
- PhD student in EMSE
- Website: [www.pingfanhu.com](http://www.pingfanhu.com)

# Meet your tutors!



## **Bogdan Bunea**

- Learning Assistant (LA)
- EMSE Junior & P4A / EDA alumni
- Check out his team's [project](#) from 2023

# Prerequisites

## EMSE 4574 / 6574: Intro to Programming for Analytics

You should be able to:

- Use RStudio to write basic R commands.
- Know the distinctions between different R operators and data types, including numeric, string, and logical data.
- Use **tidyverse** functions to wrangle and manipulate data in R.
- Use the **ggplot2** library to create plots in R.

 [Check out R for Analytics Primer](#)



# Course website

 Everything you need will be on the course website:  
<https://eda.seas.gwu.edu/2025-Fall/>

 The [schedule](#) is the best starting point

# Quizzes (10% of grade)

📅 At the start of class every other week-ish. Make ups only for excused absences (i.e. don't be late).

📅 5 total, lowest dropped

🕒 10 minutes

**Why quiz at all?** The "retrieval effect" - basically, you have to *practice* remembering things, otherwise your brain won't remember them (see the book ["Make It Stick: The Science of Successful Learning"](#))

# Assignments

- 1)  Weekly Homework / Readings: [HW1](#)
- 2)  3 Mini Projects (due 2 weeks from date assigned)
- 3)  [Final Project](#)

**Undergrads:** Teams of 3 - 4 students

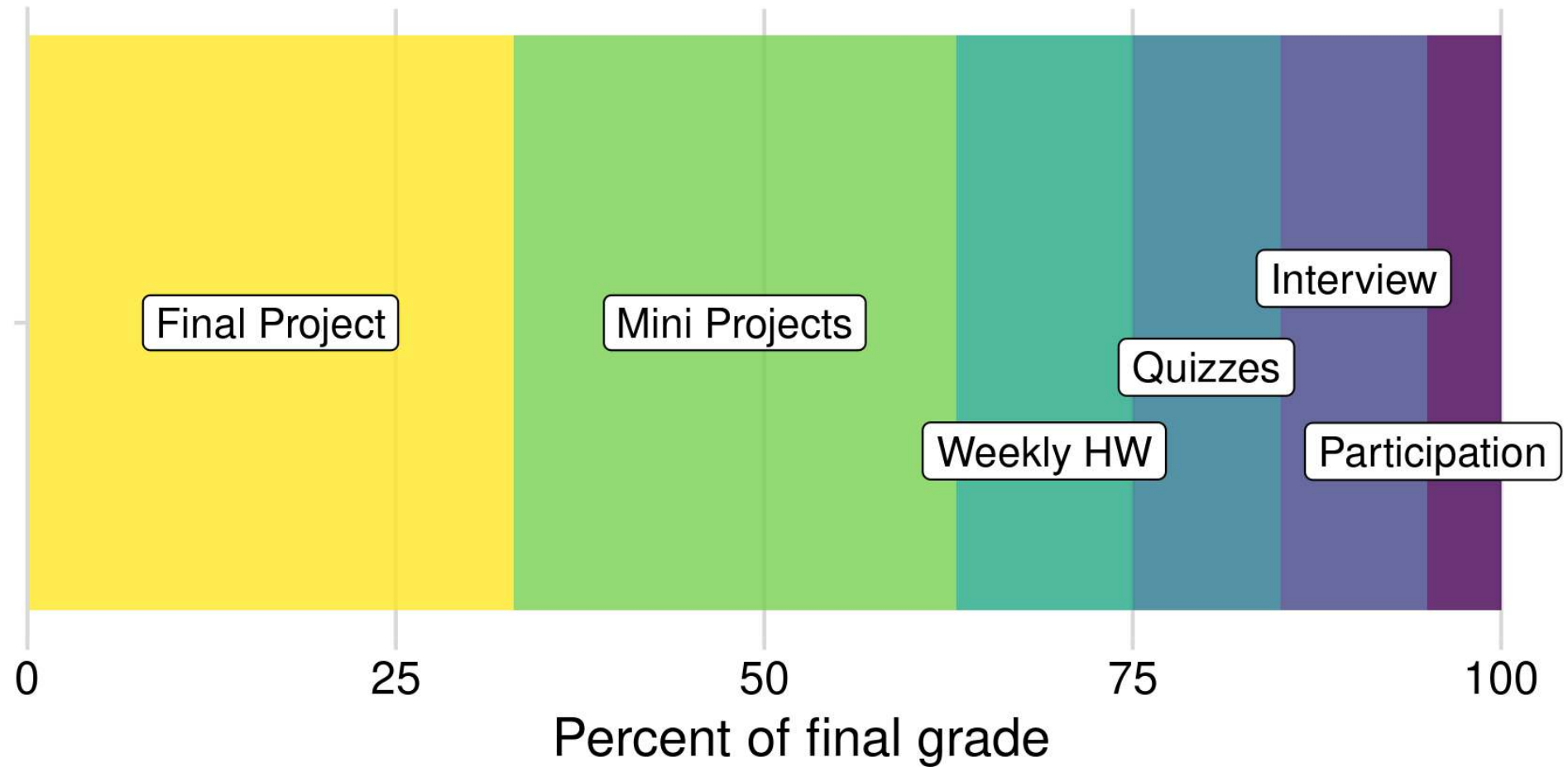
**Grads:** Teams of 2 students

| Item            | Due Date |
|-----------------|----------|
| Proposal        | Sep 21   |
| Progress Report | Oct 26   |
| Final Report    | Dec 07   |
| Presentation    | Dec 09   |

# Grades

| Item                           | Weight | Notes                              |
|--------------------------------|--------|------------------------------------|
| Participation / Attendance     | 5%     | (Yes, I take attendance)           |
| Reflections                    | 12 %   | Weekly assignment, lowest dropped) |
| Quizzes                        | 10 %   | 5 quizzes, lowest dropped          |
| Mini Project 1                 | 10 %   | Individual assignments             |
| Mini Project 2                 | 10 %   |                                    |
| Mini Project 3                 | 10 %   |                                    |
| Final Project: Proposal        | 6 %    |                                    |
| Final Project: Progress Report | 6 %    |                                    |
| Final Project: Report          | 15 %   |                                    |
| Final Project: Presentation    | 6 %    |                                    |
| Final Interview                | 10 %   | Individual interview               |

# Grades



# Course policies

- BE NICE
- BE HONEST
- DON'T CHEAT

Copying is good, stealing is bad

"Plagiarism is trying to pass someone else's work off as your own. Copying is about reverse-engineering."

-- Austin Kleon, from [Steal Like An Artist](#)

# Use of chatGPT and other AI tools

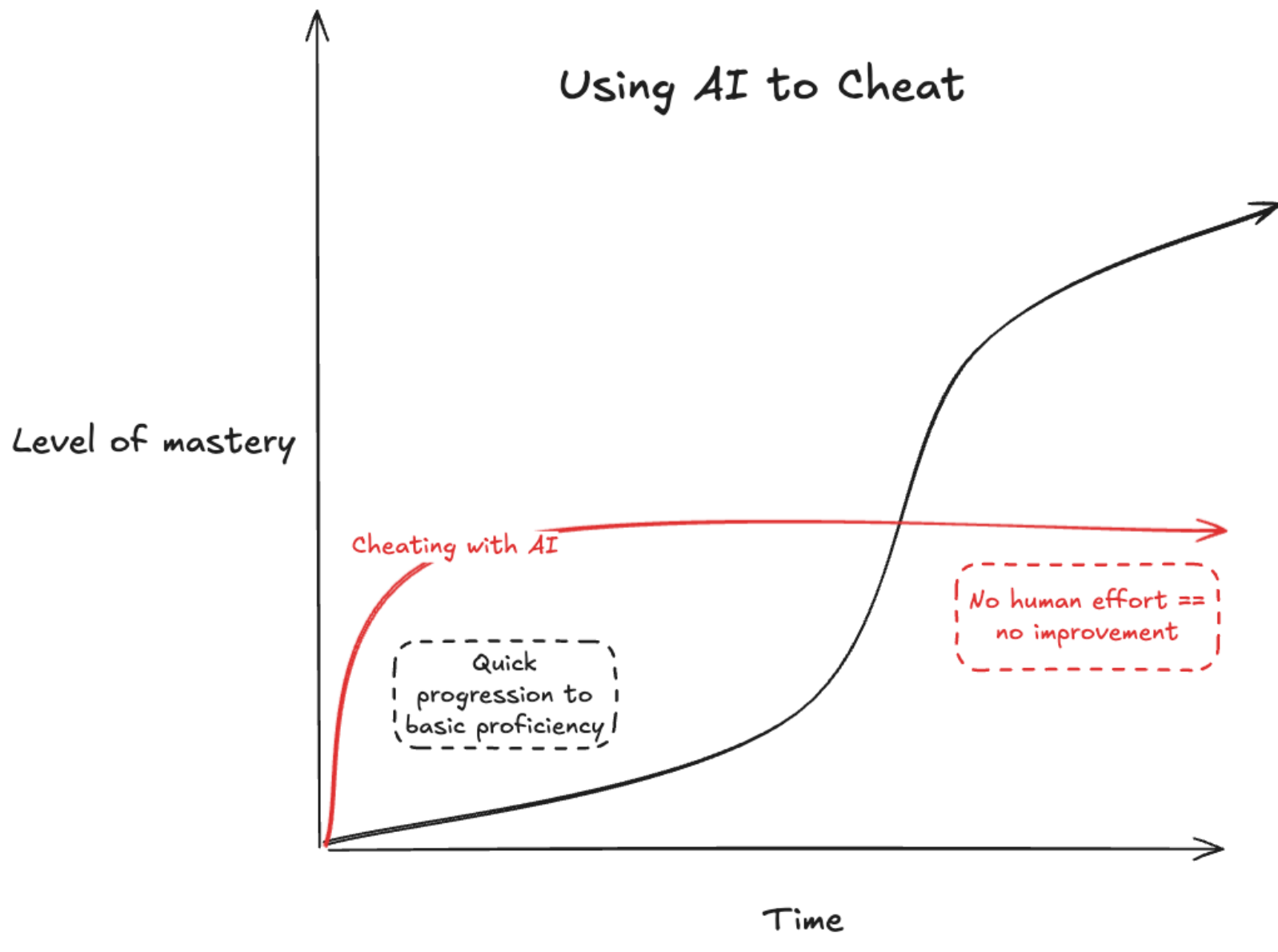
- Large language models (LLMs) are pretty good
- Sometimes they suck.
- I will grade whatever you submit. **It should not suck.**

# Ways to not have your work suck:

- Don't submit code that doesn't run (actually run it before submitting it).
- Actually read what the AI generates, and **don't submit something you don't understand**. (Ask the LLM how it works)
- There are dozens of ways to do things - **you should use the approach I teach**.
- Ask yourself "**what would I ask Prof. Helveston?**"

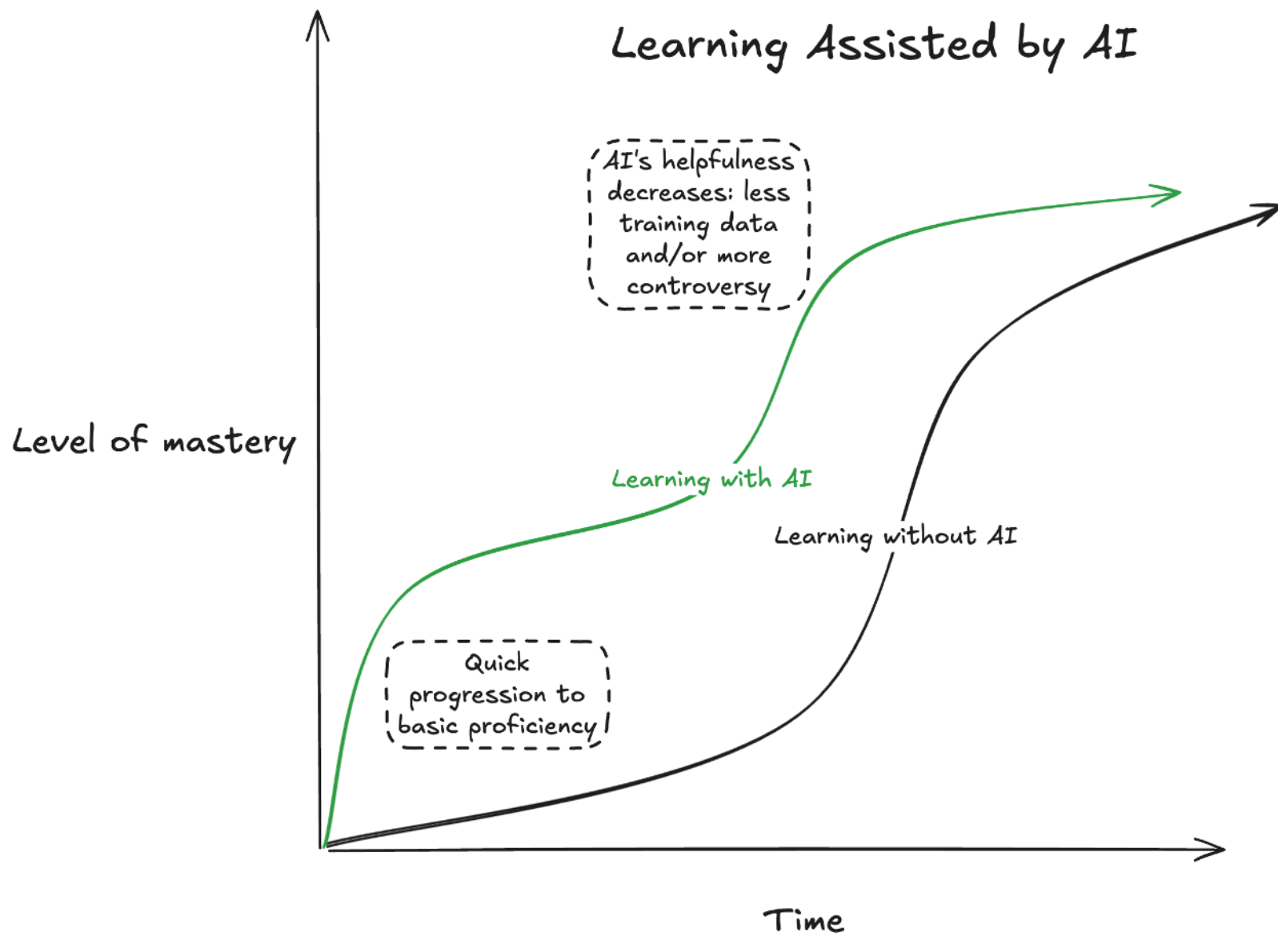
**Use AI as an assistant, not a solutions manual**





## The **wrong** ways to use LLMs

Don't copy-paste past the basics - it'll rob you of mastery



## The **right** way to use LLMs

- Asking to explain error messages
- Asking to explain *why* something works or doesn't work

# Late submissions

- **5** late days - use them anytime, no questions asked
- No more than **2** late days on any one assignment
- Contact me for special cases

# How to succeed in this class

 Participate during class!

 Start assignments early and **read carefully!**

 Actually read (before class)!

 Get sleep and take breaks often!

 Ask for help!

# Getting Help

✳️ Use [Slack](#) to ask questions.

👤💻 Meet with your tutors

👤🕒 [Schedule a meeting](#) w/ Prof. Helveston:

- Mondays from 8:00-4:30pm
- Tuesdays from 8:00-4:30pm
- Fridays from 8:00-4:00pm

</> [GW Coders](#)

## Course Software

 **Slack:** **turn notifications on!**

 **R & RStudio** (Install both)

 **Posit Cloud** (Register for free!)

# *Break*

1. If you haven't already, install everything on the [software page](#)
2. Stand up, meet each other, (maybe form teams?...use [this sheet](#))

05 : 00

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# Quick demo

1. Open `quarto_demo.qmd`
2. Click "Render"



# Anatomy of a .qmd file

Header

Markdown text

R code

# Define overall document options in header

## Basic html page

```
---  
title: Your title  
author: Author name  
format: html  
---
```

## Add table of contents, change theme

```
---  
title: Your title  
author: Author name  
toc: true  
format:  
  html:  
    theme: united  
---
```

More on themes at

<https://quarto.org/docs/output-formats/html-themes.html>

# Render to multiple outputs

## PDF uses LaTeX

```
---  
title: Your title  
author: Author name  
format: pdf  
---
```

## Microsoft Word

```
---  
title: Your title  
author: Author name  
format: docx  
---
```

If you don't have LaTeX on your computer,  
install tinytex in R:

```
tinytex::install_tinytex()
```

# Anatomy of a .qmd file

~~Header~~

Markdown text

R code

Right now, bookmark this! 📌

<https://commonmark.org/help/>

---

(When you have 10 minutes, do this! 📌)

<https://commonmark.org/help/tutorial/>

# Headers

# HEADER 1

## HEADER 2

### HEADER 3

#### HEADER 4

##### HEADER 5

##### HEADER 6

HEADER 1

HEADER 2

HEADER 3

**HEADER 4**

HEADER 5

HEADER 6

# Basic Text Formatting

Type this...

- normal text
- *\_italic text\_*
- *\*italic text\**
- **\*\*bold text\*\***
- **\*\*\*bold italic text\*\*\***
- ~~~~strikethrough~~~~
- ``code text``

..to get this

- normal text
- *italic text*
- *italic text*
- **bold text**
- ***bold italic text***
- ~~strikethrough~~
- `code text`



# Lists

Bullet list:

- first item
- second item
- third item

- first item
- second item
- third item

Numbered list:

1. first item
2. second item
3. third item

1. first item
2. second item
3. third item

# Links

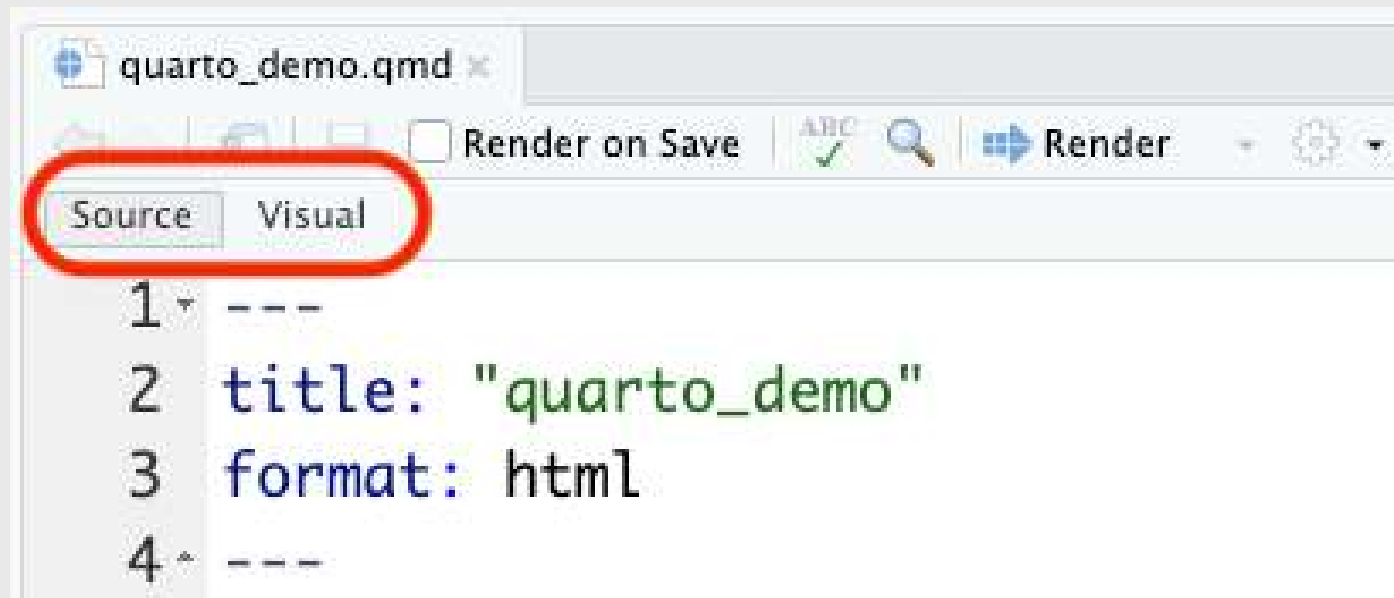
Simple **url link** to another site:

[Download R] (<http://www.r-project.org/>)

[Download R](http://www.r-project.org/)

# Don't want to use Markdown?

## Use Visual Mode!



# Anatomy of a .qmd file

~~Header (think of this as the "settings")~~

~~Markdown text~~

R code

# R Code

## Inline code

```
`r insert code here`
```

## Code chunks

```
```${r}  
insert code here  
insert more code here  
```
```

# Inline R code

```
The sum of 3 and 4 is `r 3 + 4`
```

Produces this:

The sum of 3 and 4 is 7

# R Code chunks

This code chunk...

```
```{r}
library(palmerpenguins)

head(penguins)
```
```

...will produce this when compiled:

```
library(palmerpenguins)

head(penguins)
```

```
#> # A tibble: 6 × 8
#>   species island    bill_length_mm bill_c
#>   <fct>    <fct>          <dbl>
#> 1 Adelie  Torgersen         39.1
#> 2 Adelie  Torgersen         39.5
#> 3 Adelie  Torgersen         40.3
#> 4 Adelie  Torgersen          NA
#> 5 Adelie  Torgersen         36.7
#> 6 Adelie  Torgersen         39.3
```

# Chunk options

Control what chunks output using options

All options [here](#)

| option     | default  | effect                                                    |
|------------|----------|-----------------------------------------------------------|
| eval       | TRUE     | Whether to evaluate the code and include its results      |
| echo       | TRUE     | Whether to display code along with its results            |
| warning    | TRUE     | Whether to display warnings                               |
| error      | FALSE    | Whether to display errors                                 |
| message    | TRUE     | Whether to display messages                               |
| tidy       | FALSE    | Whether to reformat code in a tidy way when displaying it |
| results    | "markup" | "markup", "asis", "hold", or "hide"                       |
| cache      | FALSE    | Whether to cache results for future renders               |
| comment    | "##"     | Comment character to preface results with                 |
| fig.width  | 7        | Width in inches for plots created in chunk                |
| fig.height | 7        | Height in inches for plots created in chunk               |



# Chunk output options

By default, code chunks print **code** + **output**

```
```${r}  
#| echo: false  
  
cat('hello world!')  
```
```

Prints only **output**  
(doesn't show code)

```
#> hello world!
```

```
```${r}  
#| eval: false  
  
cat('hello world!')  
```
```

Prints only **code**  
(doesn't run the code)

```
cat('hello world!')
```

```
```${r}  
#| include: false  
  
cat('hello world!')  
```
```

Runs, but doesn't print  
anything

# A global `setup` chunk 🌍

```
```\{r}  
#| label: setup  
#| include: false  
  
knitr::opts_chunk$set(  
  warning = FALSE,  
  message = FALSE,  
  fig.path = "figs/",  
  fig.width = 7.252,  
  fig.height = 4,  
  comment = "#>",  
  fig.retina = 3  
)  
```
```

- Typically the first chunk
- All following chunks will use these options (i.e., sets global chunk options)
- You can (and should) use individual chunk options too
- Often where I load libraries, etc.

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# Workflow for reading in data

1) Use R Projects (.Rproj files) to organize your analysis - **don't double-click .R files!**



2) Use the [here](#) package to create file paths

```
path <- here::here("folder", "file.csv")
```

3) Import data with these functions:

| File type | Function     | Library       |
|-----------|--------------|---------------|
| .csv      | read_csv()   | <b>readr</b>  |
| .txt      | read_table() | <b>utils</b>  |
| .xlsx     | read_excel() | <b>readxl</b> |

# Importing Comma Separated Values (.csv)

Read in `.csv` files with `read_csv()`:

```
library(tidyverse)
library(here)

csvPath <- here('data', 'milk_production.csv')
milk_production <- read_csv(csvPath)

head(milk_production)
```

```
#> # A tibble: 6 × 4
#>   region    state    year milk_produced
#>   <chr>    <chr>    <dbl>         <dbl>
#> 1 Northeast Maine      1970      619000000
#> 2 Northeast New Hampshire 1970      356000000
#> 3 Northeast Vermont    1970     1970000000
#> 4 Northeast Massachusetts 1970      658000000
#> 5 Northeast Rhode Island 1970      75000000
#> 6 Northeast Connecticut 1970     661000000
```

# Importing Text Files (.txt)

Read in `.txt` files with `read.table()`:

```
txtPath <- here('data', 'nasa_global_temps.txt')
global_temps <- read.table(txtPath, skip = 5, header = FALSE)
head(global_temps)
```

```
#>      V1      V2      V3
#> 1 1880 -0.15 -0.08
#> 2 1881 -0.07 -0.12
#> 3 1882 -0.10 -0.15
#> 4 1883 -0.16 -0.19
#> 5 1884 -0.27 -0.23
#> 6 1885 -0.32 -0.25
```

# Importing Text Files (.txt)

Read in `.txt` files with `read.table()`:

```
txtPath <- here('data', 'nasa_global_temps.txt')
global_temps <- read.table(txtPath, skip = 5, header = FALSE)
names(global_temps) <- c('year', 'no_smoothing', 'loess') # Add header
head(global_temps)
```

```
#>   year no_smoothing loess
#> 1 1880         -0.15 -0.08
#> 2 1881         -0.07 -0.12
#> 3 1882         -0.10 -0.15
#> 4 1883         -0.16 -0.19
#> 5 1884         -0.27 -0.23
#> 6 1885         -0.32 -0.25
```

# Importing Excel Files (.xlsx)

Read in `.xlsx` files with `read_excel()`:

```
library(readxl)
```

```
xlsxPath <- here('data', 'pv_cell_production.xlsx')
```

```
pv_cells <- read_excel(xlsxPath, sheet = 'Cell Prod by Country', skip = 2)
```

```
glimpse(pv_cells)
```

```
#> Rows: 25
#> Columns: 10
#> $ Year      <chr> NA, NA, "1995", "1996", "1997", "1998", "1999", "2000", "2001", "2002"
#> $ China     <chr> "Megawatts", NA, "NA", "NA", "NA", "NA", "NA", "NA", "2.5", "3", "10", "13"
#> $ Taiwan    <chr> NA, NA, "NA", "NA", "NA", "NA", "NA", "NA", "3.5", "8", "17", "39.299"
#> $ Japan     <dbl> NA, NA, 16.4, 21.2, 35.0, 49.0, 80.0, 128.6, 171.2, 251.1, 363.9, 601
#> $ Malaysia  <chr> NA, NA, "NA", "NA", "NA", "NA", "NA", "NA", "0", "0", "0", "0", "0",
#> $ Germany   <chr> NA, NA, "NA", "NA", "NA", "NA", "NA", "NA", "22.5", "23.5", "55", "121.5",
#> $ `South Korea` <chr> NA, NA, "NA", "NA", "NA", "NA", "NA", "NA", "0", "0", "0", "0", "5.3"
#> $ `United States` <dbl> NA, NA, 34.7500, 38.8500, 51.0000, 53.7000, 60.8000, 75.0000, 100.300
#> $ Others    <chr> NA, NA, "NA", "NA", "NA", "NA", "NA", "NA", "48.200000000000017", "69.80000
#> $ World     <dbl> NA, NA, 77.600, 88.600, 125.800, 154.900, 201.300, 276.800, 371.300,
```



# Importing Excel Files (.xlsx)

Read in `.xlsx` files with `read_excel()`:

```
library(readxl)

xlsxPath <- here('data', 'pv_cell_production.xlsx')
pv_cells <- read_excel(xlsxPath, sheet = 'Cell Prod by Country', skip = 2) %>%
  mutate(Year = as.numeric(Year)) %>% # Convert "non-years" to NA
  filter(!is.na(Year)) # Drop NA rows in Year
```

```
glimpse(pv_cells)
```

```
#> Rows: 19
#> Columns: 10
#> $ Year      <dbl> 1995, 1996, 1997, 1998, 1999, 2000, 2001, 2002, 2003, 2004, 2005, 2006, 2007, 2008, 20
#> $ China     <chr> "NA", "NA", "NA", "NA", "NA", "2.5", "3", "10", "13", "40", "128.30000000000001", "34
#> $ Taiwan    <chr> "NA", "NA", "NA", "NA", "NA", "NA", "3.5", "8", "17", "39.299999999999997", "88", "16
#> $ Japan     <dbl> 16.4, 21.2, 35.0, 49.0, 80.0, 128.6, 171.2, 251.1, 363.9, 601.5, 833.0, 926.4, 937.5,
#> $ Malaysia  <chr> "NA", "NA", "NA", "NA", "NA", "NA", "0", "0", "0", "0", "0", "0", "100.1", "397.9", "
#> $ Germany   <chr> "NA", "NA", "NA", "NA", "NA", "22.5", "23.5", "55", "121.5", "193", "339", "469.1", "
#> $ `South Korea` <chr> "NA", "NA", "NA", "NA", "NA", "NA", "0", "0", "0", "0", "5.3", "13", "31.883935905674
#> $ `United States` <dbl> 34.7500, 38.8500, 51.0000, 53.7000, 60.8000, 75.0000, 100.3000, 120.6000, 103.0000, 1
#> $ Others    <chr> "NA", "NA", "NA", "NA", "NA", "48.200000000000017", "69.800000000000011", "97.2999999
#> $ World     <dbl> 77.600, 88.600, 125.800, 154.900, 201.300, 276.800, 371.300, 542.000, 749.400, 1198.80
```

# Your turn

10:00

Open the `practice.qmd` file.

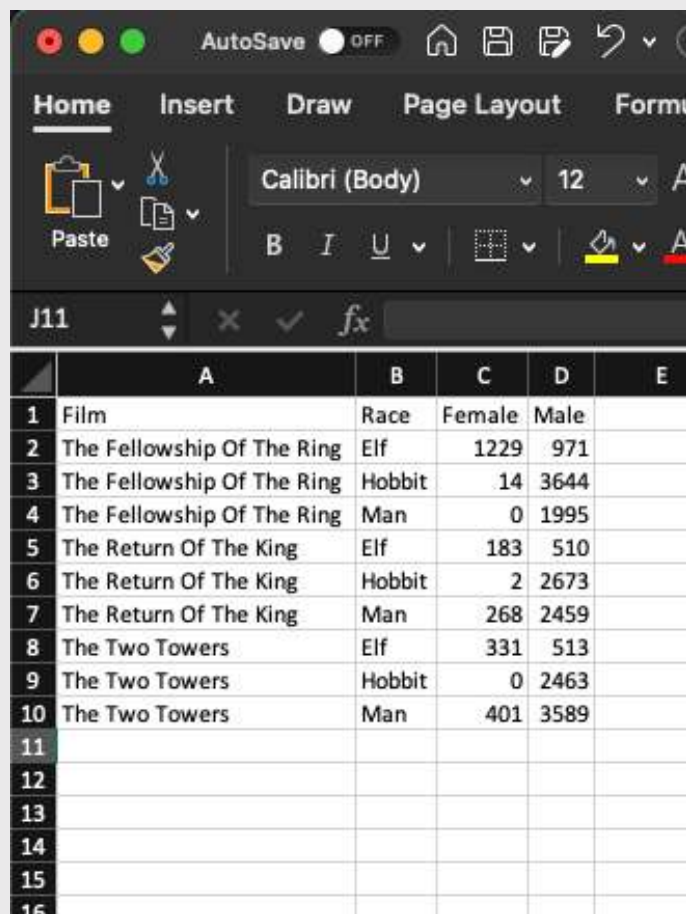
Write code to import the following data files from the "data" folder:

- For `lotr_words.csv`, call the data frame `lotr`
- For `north_america_bear_killings.txt`, call the data frame `bears`
- For `uspto_clean_energy_patents.xlsx`, call the data frame `patents`

# Week 1: *Getting Started*

1. Course Goal
2. Course Introduction
3. Break: Install Stuff
4. Quarto
5. Workflow & Reading In Data
6. **Wrangling Data**
7. Visualizing Data

# The data frame... in Excel



A screenshot of the Microsoft Excel interface. The 'Home' tab is selected, showing the 'Paste' button and font settings (Calibri, size 12). The data is organized in a table with columns A through E. Row 1 contains headers: 'Film', 'Race', 'Female', and 'Male'. Rows 2 through 10 contain movie data.

|    | A                          | B      | C      | D    | E |
|----|----------------------------|--------|--------|------|---|
| 1  | Film                       | Race   | Female | Male |   |
| 2  | The Fellowship Of The Ring | Elf    | 1229   | 971  |   |
| 3  | The Fellowship Of The Ring | Hobbit | 14     | 3644 |   |
| 4  | The Fellowship Of The Ring | Man    | 0      | 1995 |   |
| 5  | The Return Of The King     | Elf    | 183    | 510  |   |
| 6  | The Return Of The King     | Hobbit | 2      | 2673 |   |
| 7  | The Return Of The King     | Man    | 268    | 2459 |   |
| 8  | The Two Towers             | Elf    | 331    | 513  |   |
| 9  | The Two Towers             | Hobbit | 0      | 2463 |   |
| 10 | The Two Towers             | Man    | 401    | 3589 |   |
| 11 |                            |        |        |      |   |
| 12 |                            |        |        |      |   |
| 13 |                            |        |        |      |   |
| 14 |                            |        |        |      |   |
| 15 |                            |        |        |      |   |
| 16 |                            |        |        |      |   |

# The data frame... in R

lotr

```
#> # A tibble: 18 × 4
#>   film                                race  gender
#>   <chr>                                <chr>  <chr>
#> 1 The Fellowship Of The Ring  Elf    Female
#> 2 The Fellowship Of The Ring  Elf    Male
#> 3 The Fellowship Of The Ring  Hobbit Female
#> 4 The Fellowship Of The Ring  Hobbit Male
#> 5 The Fellowship Of The Ring  Man    Female
#> 6 The Fellowship Of The Ring  Man    Male
#> 7 The Return Of The King      Elf    Female
#> 8 The Return Of The King      Elf    Male
#> 9 The Return Of The King      Hobbit Female
#> 10 The Return Of The King      Hobbit Male
#> 11 The Return Of The King      Man    Female
#> 12 The Return Of The King      Man    Male
#> 13 The Two Towers             Elf    Female
#> 14 The Two Towers             Elf    Male
```

# Columns: *Vectors* of values (must be same data type)

Extract a column using `$`

```
lotr$race
```

```
#> [1] "Elf"      "Elf"      "Hobbit"   "Hobbit"   "Man"      "Man"      "Elf"      "Elf"      "Hobbit"   "
```

# Columns: *Vectors* of values (must be same data type)

Can also use brackets:

```
lotr$race
```

```
#> [1] "Elf" "Elf" "Hobbit" "Hobbit" "Man" "Man" "Elf" "Elf" "Hobbit" "
```

```
lotr[,2]
```

```
#> # A tibble: 18 × 1  
#>   race  
#>   <chr>  
#> 1 Elf  
#> 2 Elf  
#> 3 Hobbit  
#> 4 Hobbit  
#> 5 Man  
#> 6 Man  
#> 7 Elf  
#> 8 Elf  
#> 9 Hobbit
```

# Rows: Information about individual observations

Information about the first row:

```
lotr[1,]
```

```
#> # A tibble: 1 × 4  
#>   film          race gender word_count  
#>   <chr>        <chr> <chr>     <dbl>  
#> 1 The Fellowship Of The Ring Elf      Female      1229
```

Information about rows 1 & 2:

```
lotr[1:2,]
```

```
#> # A tibble: 2 × 4  
#>   film          race gender word_count  
#>   <chr>        <chr> <chr>     <dbl>  
#> 1 The Fellowship Of The Ring Elf      Female      1229  
#> 2 The Fellowship Of The Ring Elf      Male        971
```

# Quick Practice

Read in the `data.csv` file in the "data" folder:

```
data <- read_csv(here('data', 'data.csv'))
```

Now answer these questions:

- How many rows and columns are in the data frame?
- What type of data is each column?
- Preview the different columns - what do you think this data is about? What might one row represent?
- How many unique airlines are in the data frame?
- What is the shortest and longest air time for any one flight in the data frame?



The tidyverse: `stringr` + `dplyr` + `readr` + `ggplot2` + ...



Art by [Allison Horst](#)

# The main `dplyr` "verbs"

| "Verb"                   | What it does                  |
|--------------------------|-------------------------------|
| <code>select()</code>    | Select columns by name        |
| <code>filter()</code>    | Keep rows that match criteria |
| <code>arrange()</code>   | Sort rows based on column(s)  |
| <code>mutate()</code>    | Create new columns            |
| <code>summarize()</code> | Create summary values         |

# Core `tidyverse` concept: **Chain functions together with "pipes"**

`%>%`

Think of the words "...and then..."

```
data %>%  
  do_something() %>%  
  do_something_else()
```

Select columns with `select()`

## Subset Variables (Columns)



# Select columns with `select()`

Select the columns `film` & `race`

```
lotr %>%  
  select(film, race)
```

```
#> # A tibble: 18 × 2  
#>   film                                race  
#>   <chr>                             <chr>  
#> 1 The Fellowship Of The Ring      Elf  
#> 2 The Fellowship Of The Ring      Elf  
#> 3 The Fellowship Of The Ring      Hobbit  
#> 4 The Fellowship Of The Ring      Hobbit  
#> 5 The Fellowship Of The Ring      Man  
#> 6 The Fellowship Of The Ring      Man  
#> 7 The Return Of The King           Elf  
#> 8 The Return Of The King           Elf  
#> 9 The Return Of The King           Hobbit  
#> 10 The Return Of The King          Hobbit  
#> 11 The Return Of The King          Man  
#> 12 The Return Of The King          Man  
#> 13 The Two Towers                  Elf
```

# Select columns with `select()`

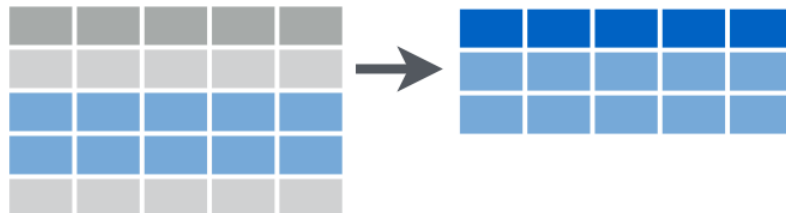
Use the `-` sign to drop columns

```
lotr %>%  
  select(-film)
```

```
#> # A tibble: 18 × 3  
#>   race    gender word_count  
#>   <chr>  <chr>      <dbl>  
#> 1 Elf    Female      1229  
#> 2 Elf    Male        971  
#> 3 Hobbit Female        14  
#> 4 Hobbit Male      3644  
#> 5 Man    Female         0  
#> 6 Man    Male      1995  
#> 7 Elf    Female       183  
#> 8 Elf    Male       510  
#> 9 Hobbit Female         2  
#> 10 Hobbit Male     2673  
#> 11 Man    Female       268  
#> 12 Man    Male     2459  
#> 13 Elf    Female       231
```

# Filter for rows with `filter()`

## Subset Observations (Rows)



# Filter for rows with `filter()`

Keep only the rows with Elf characters

```
lotr %>%  
  filter(race == "Elf")
```

```
#> # A tibble: 6 × 4  
#>   film      race gender word_count  
#>   <chr>   <chr> <chr>      <dbl>  
#> 1 The Fellowship Of The Ring Elf    Female    1229  
#> 2 The Fellowship Of The Ring Elf    Male      971  
#> 3 The Return Of The King    Elf    Female    183  
#> 4 The Return Of The King    Elf    Male     510  
#> 5 The Two Towers            Elf    Female    331  
#> 6 The Two Towers            Elf    Male     513
```



# Filter for rows with `filter()`

Keep only the rows with Elf or Hobbit characters

```
lotr %>%  
  filter((race == "Elf") | (race == "Hobbit"))
```

```
#> # A tibble: 12 × 4  
#>   film      race gender word_count  
#>   <chr>    <chr>  <chr>      <dbl>  
#> 1 The Fellowship Of The Ring Elf      Female    1229  
#> 2 The Fellowship Of The Ring Elf      Male      971  
#> 3 The Fellowship Of The Ring Hobbit    Female     14  
#> 4 The Fellowship Of The Ring Hobbit    Male    3644  
#> 5 The Return Of The King    Elf      Female    183  
#> 6 The Return Of The King    Elf      Male     510  
#> 7 The Return Of The King    Hobbit    Female      2  
#> 8 The Return Of The King    Hobbit    Male    2673  
#> 9 The Two Towers           Elf      Female    331  
#> 10 The Two Towers           Elf      Male     513  
#> 11 The Two Towers           Hobbit    Female      0  
#> 12 The Two Towers           Hobbit    Male    2463
```

# Filter for rows with `filter()`

Keep only the rows with Elf or Hobbit characters

```
lotr %>%  
  filter(race %in% c("Elf", "Hobbit"))
```

```
#> # A tibble: 12 × 4  
#>   film      race gender word_count  
#>   <chr>    <chr>  <chr>      <dbl>  
#> 1 The Fellowship Of The Ring Elf      Female    1229  
#> 2 The Fellowship Of The Ring Elf      Male      971  
#> 3 The Fellowship Of The Ring Hobbit    Female     14  
#> 4 The Fellowship Of The Ring Hobbit    Male    3644  
#> 5 The Return Of The King    Elf      Female    183  
#> 6 The Return Of The King    Elf      Male     510  
#> 7 The Return Of The King    Hobbit    Female      2  
#> 8 The Return Of The King    Hobbit    Male    2673  
#> 9 The Two Towers           Elf      Female    331  
#> 10 The Two Towers           Elf      Male     513  
#> 11 The Two Towers           Hobbit    Female      0  
#> 12 The Two Towers           Hobbit    Male    2463
```

# Logic operators for `filter()`

| Description                            | Example                         |
|----------------------------------------|---------------------------------|
| Values greater than 1                  | <code>value &gt; 1</code>       |
| Values greater than or equal to 1      | <code>value &gt;= 1</code>      |
| Values less than 1                     | <code>value &lt; 1</code>       |
| Values less than or equal to 1         | <code>value &lt;= 1</code>      |
| Values equal to 1                      | <code>value == 1</code>         |
| Values not equal to 1                  | <code>value != 1</code>         |
| Values in the set <code>c(1, 4)</code> | <code>value %in% c(1, 4)</code> |

# Combine `filter()` and `select()`

Keep only the rows with Elf characters that spoke more than 1000 words, then select everything but the race column

```
lotr %>%  
  filter((race == "Elf") & (word_count > 1000)) %>%  
  select(-race)
```

```
#> # A tibble: 1 × 3  
#>   film                gender word_count  
#>   <chr>              <chr>      <dbl>  
#> 1 The Fellowship Of The Ring Female      1229
```

Create new variables with `mutate()`

## Make New Variables



# Create new variables with `mutate()`

Create a new variable, `word1000` which is `TRUE` if the character spoke 1,000 or more words

```
lotr %>%  
  mutate(word1000 = word_count >= 1000)
```

```
#> # A tibble: 18 × 5  
#>   film      race gender word_count word1000  
#>   <chr>    <chr> <chr>      <dbl> <lgl>  
#> 1 The Fellowship Of The Ring Elf      Female      1229 TRUE  
#> 2 The Fellowship Of The Ring Elf      Male        971 FALSE  
#> 3 The Fellowship Of The Ring Hobbit    Female        14 FALSE  
#> 4 The Fellowship Of The Ring Hobbit    Male      3644 TRUE  
#> 5 The Fellowship Of The Ring Man       Female        0 FALSE  
#> 6 The Fellowship Of The Ring Man       Male      1995 TRUE  
#> 7 The Return Of The King    Elf      Female      183 FALSE  
#> 8 The Return Of The King    Elf      Male       510 FALSE  
#> 9 The Return Of The King    Hobbit   Female        2 FALSE  
#> 10 The Return Of The King   Hobbit   Male     2673 TRUE  
#> 11 The Return Of The King   Man      Female      268 FALSE
```

# Handling if/else conditions

`ifelse(<condition>, <if TRUE>, <else>)`

```
lotr %>%  
  mutate(word1000 = ifelse(word_count >= 1000, TRUE, FALSE))
```

```
#> # A tibble: 18 × 5  
#>   film      race gender word_count word1000  
#>   <chr>    <chr>  <chr>    <dbl> <lgl>  
#> 1 The Fellowship Of The Ring Elf      Female    1229 TRUE  
#> 2 The Fellowship Of The Ring Elf      Male      971 FALSE  
#> 3 The Fellowship Of The Ring Hobbit   Female     14 FALSE  
#> 4 The Fellowship Of The Ring Hobbit   Male    3644 TRUE  
#> 5 The Fellowship Of The Ring Man      Female     0 FALSE  
#> 6 The Fellowship Of The Ring Man      Male    1995 TRUE  
#> 7 The Return Of The King    Elf      Female    183 FALSE  
#> 8 The Return Of The King    Elf      Male     510 FALSE  
#> 9 The Return Of The King    Hobbit   Female     2 FALSE  
#> 10 The Return Of The King    Hobbit   Male    2673 TRUE  
#> 11 The Return Of The King    Man      Female    268 FALSE  
#> 12 The Return Of The King    Man      Male    2459 TRUE
```

# Sort data frame with `arrange()`

Sort the `lotr` data frame by `word_count`

```
lotr %>%  
  arrange(word_count)
```

```
#> # A tibble: 18 × 4  
#>   film          race gender word_count  
#>   <chr>         <chr> <chr>      <dbl>  
#> 1 The Fellowship Of The Ring Man    Female      0  
#> 2 The Two Towers Hobbit Female      0  
#> 3 The Return Of The King Hobbit Female      2  
#> 4 The Fellowship Of The Ring Hobbit Female     14  
#> 5 The Return Of The King Elf     Female    183  
#> 6 The Return Of The King Man     Female    268  
#> 7 The Two Towers Elf     Female    331  
#> 8 The Two Towers Man     Female    401  
#> 9 The Return Of The King Elf     Male     510  
#> 10 The Two Towers Elf     Male     513  
#> 11 The Fellowship Of The Ring Elf     Male     971  
#> 12 The Fellowship Of The Ring Elf     Female   1229  
#> 13 The Fellowship Of The Ring Man     Male     1005
```



# Sort data frame with `arrange()`

Use the `desc()` function to sort in descending order

```
lotr %>%  
  arrange(desc(word_count))
```

```
#> # A tibble: 18 × 4  
#>   film          race gender word_count  
#>   <chr>         <chr> <chr>      <dbl>  
#> 1 The Fellowship Of The Ring Hobbit Male      3644  
#> 2 The Two Towers      Man   Male      3589  
#> 3 The Return Of The King Hobbit Male      2673  
#> 4 The Two Towers      Hobbit Male      2463  
#> 5 The Return Of The King Man    Male      2459  
#> 6 The Fellowship Of The Ring Man    Male      1995  
#> 7 The Fellowship Of The Ring Elf    Female    1229  
#> 8 The Fellowship Of The Ring Elf    Male      971  
#> 9 The Two Towers      Elf    Male      513  
#> 10 The Return Of The King Elf    Male      510  
#> 11 The Two Towers      Man    Female     401  
#> 12 The Two Towers      Elf    Female     331  
#> 13 The Return Of The King Man    Female     268
```

# Your turn

10:00

Read in the `data.csv` file in the "data" folder:

```
data <- read_csv(here('data', 'data.csv'))
```

Now answer these questions:

- Create a new data frame, `flights_fall`, that contains only flights that departed in the fall semester.
- Create a new data frame, `flights_dc`, that contains only flights that flew to DC airports (Reagan or Dulles).
- Create a new data frame, `flights_dc_carrier`, that contains only flights that flew to DC airports (Reagan or Dulles) and only the columns about the month and airline.
- How many unique airlines were flying to DC airports in July?
- Create a new variable, `speed`, in miles per hour using the `time` (minutes) and `distance` (miles) variables.
- Which flight flew the fastest?
- Remove rows that have `NA` for `air_time` and re-arrange the resulting data frame based on the longest air time and longest flight distance.

# Week 1: *Getting Started*

1. Course Goal
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7. Visualizing Data

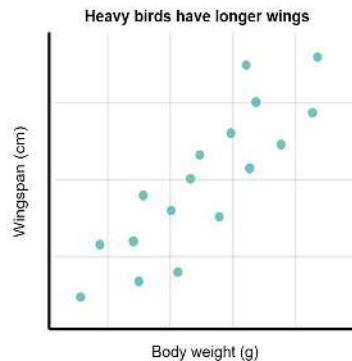
## MAKING A GRAPH WITH GGPLOT2

Customise the look of your plot with themes  
(pre-made or your own!):  
`+ theme_bw()`

Add labels and titles:  
`+ labs(x = "Body weight (g)", y = "Wingspan (cm)",  
title = "Heavy birds have longer wings")`

Specify the type of graph and the variables to use:  
`+ geom_point(aes(x = body.weight, y = wingspan))`

Plot the device containing your data:  
`ggplot(data = birds)`



# "Grammar of Graphics"

Concept developed by Leland Wilkinson  
(1999)

**ggplot2** package developed by Hadley  
Wickham (2005)

# Making plot layers with ggplot2

1. The data
2. The aesthetic mapping (what goes on the axes?)
3. The geometries (points? bars? etc.)
4. The annotations / labels
5. The theme

# Layer 1: The data

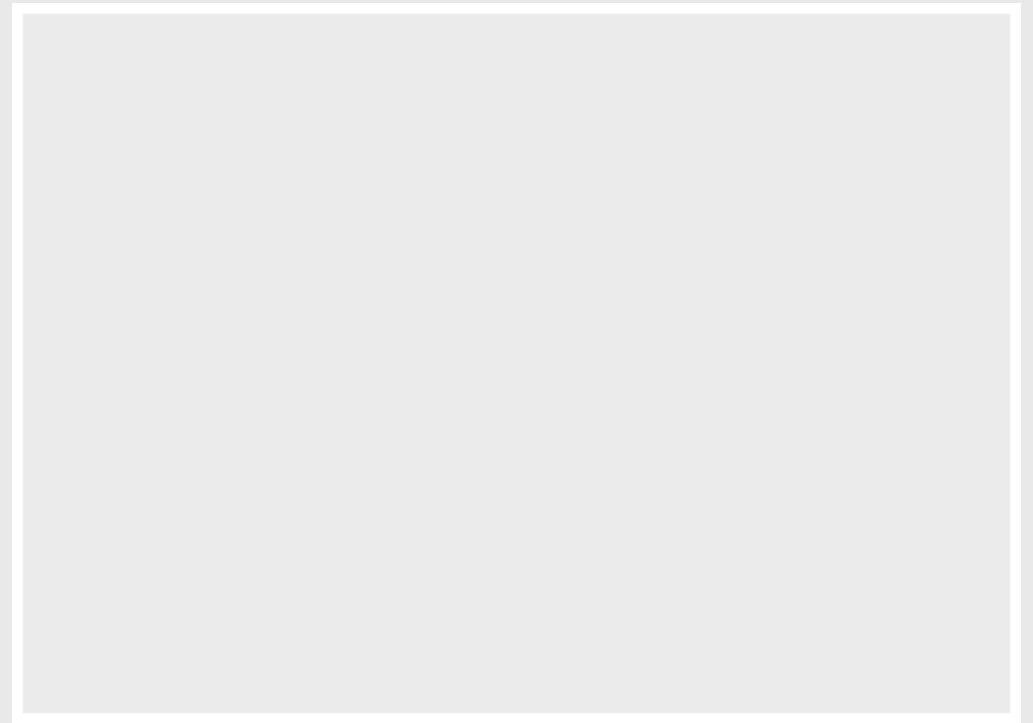
```
head(mpg)
```

```
#> # A tibble: 6 × 11
#>   manufacturer model displ  year   cyl trans      drv    cty   hwy fl      class
#>   <chr>         <chr> <dbl> <int> <int> <chr>    <chr> <int> <int> <chr> <chr>
#> 1 audi         a4      1.8  1999     4 auto(l5)  f      18    29 p      compact
#> 2 audi         a4      1.8  1999     4 manual(m5) f      21    29 p      compact
#> 3 audi         a4      2    2008     4 manual(m6) f      20    31 p      compact
#> 4 audi         a4      2    2008     4 auto(av)   f      21    30 p      compact
#> 5 audi         a4      2.8  1999     6 auto(l5)  f      16    26 p      compact
#> 6 audi         a4      2.8  1999     6 manual(m5) f      18    26 p      compact
```

# Layer 1: The data

The `ggplot()` function initializes the plot with whatever data you're using

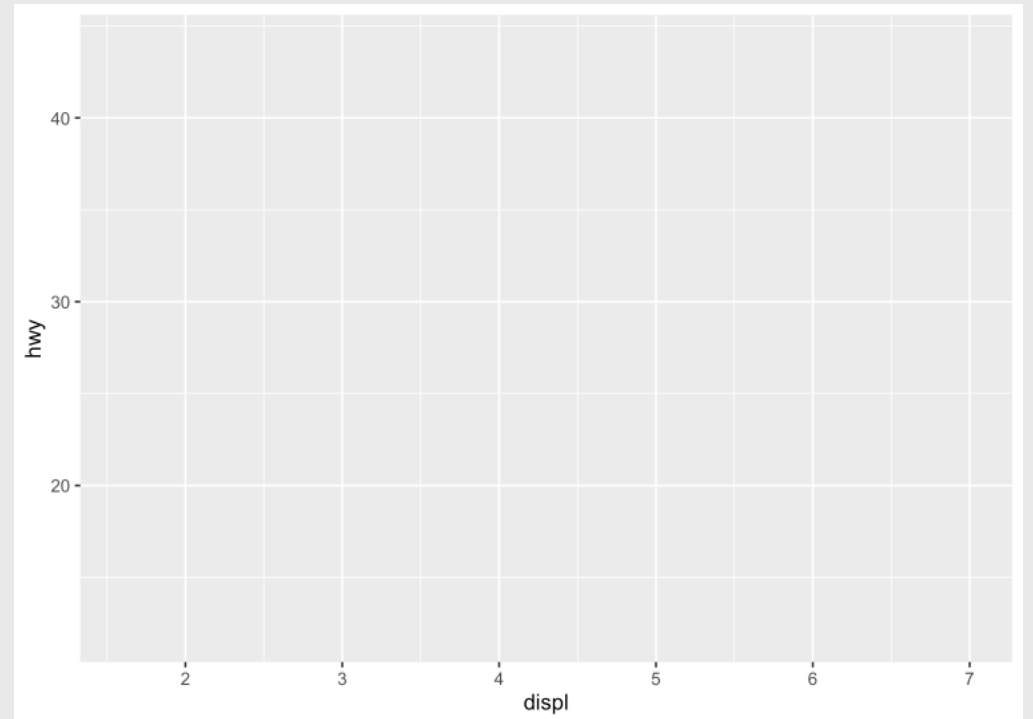
```
mpg %>%  
  ggplot()
```



# Layer 2: The aesthetic mapping

The `aes()` function determines which variables will be *mapped* to the geometries (e.g. the axes)

```
mpg %>%  
  ggplot(aes(x = displ, y = hwy))
```

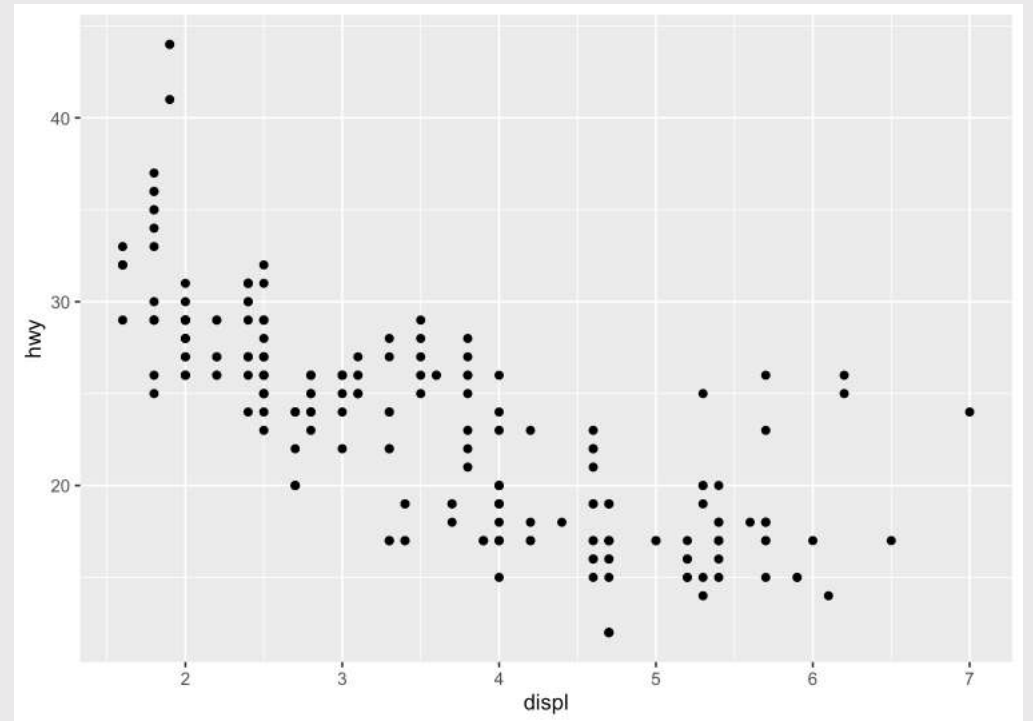




# Layer 3: The geometries

Use `+` to add geometries, e.g. `geom_points()` for points

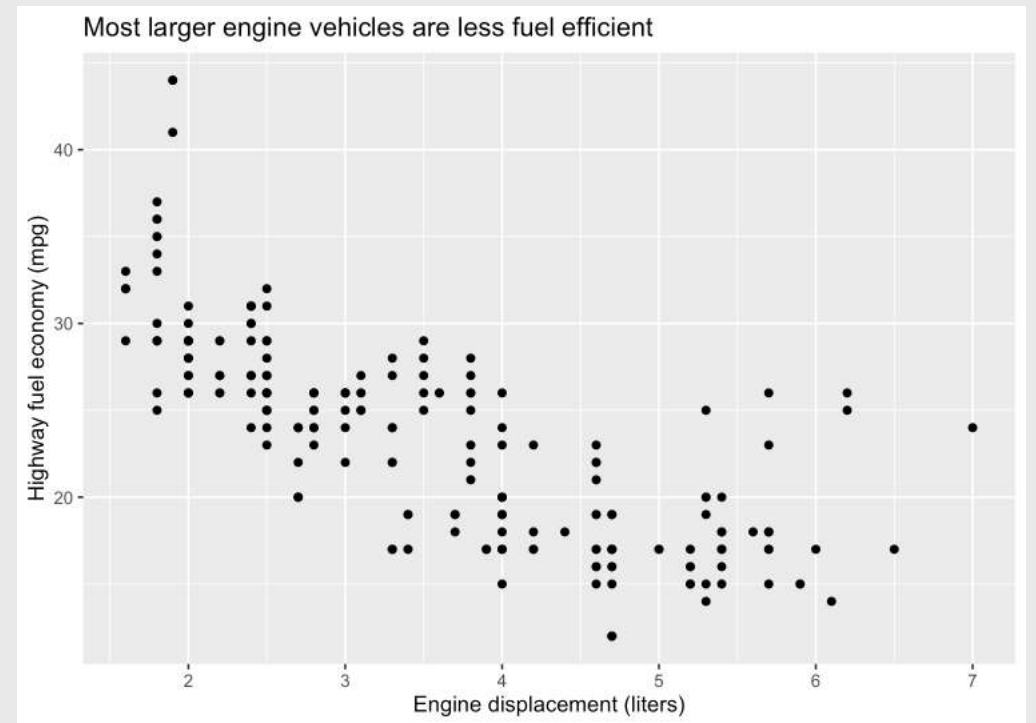
```
mpg %>%  
  ggplot(aes(x = displ, y = hwy)) +  
  geom_point()
```



# Layer 4: The annotations / labels

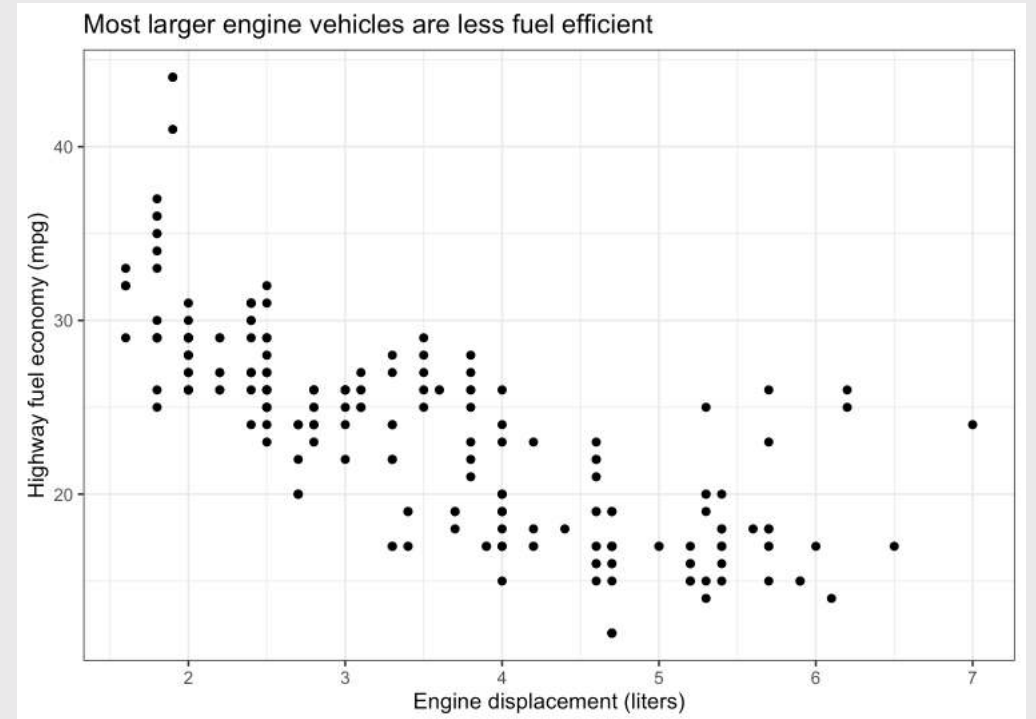
Use `labs()` to modify most labels

```
mpg %>%  
  ggplot(aes(x = displ, y = hwy)) +  
  geom_point() +  
  labs(  
    x = "Engine displacement (liters)",  
    y = "Highway fuel economy (mpg)",  
    title = "Most larger engine vehicles are less fuel efficient"  
  )
```



# Layer 5: The theme

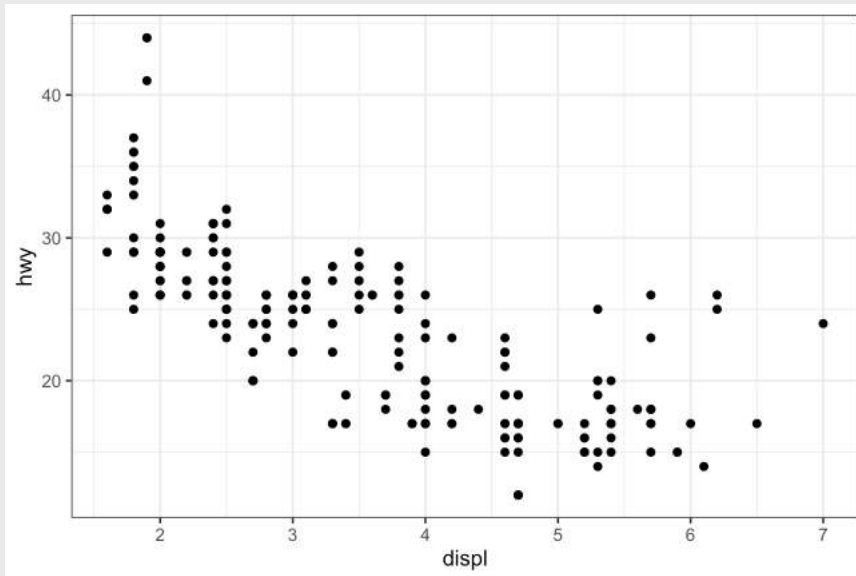
```
mpg %>%  
  ggplot(aes(x = displ, y = hwy)) +  
  geom_point() +  
  labs(  
    x = "Engine displacement (liters)",  
    y = "Highway fuel economy (mpg)",  
    title = "Most larger engine vehicles are  
  ) +  
  theme_bw()
```



# Common themes

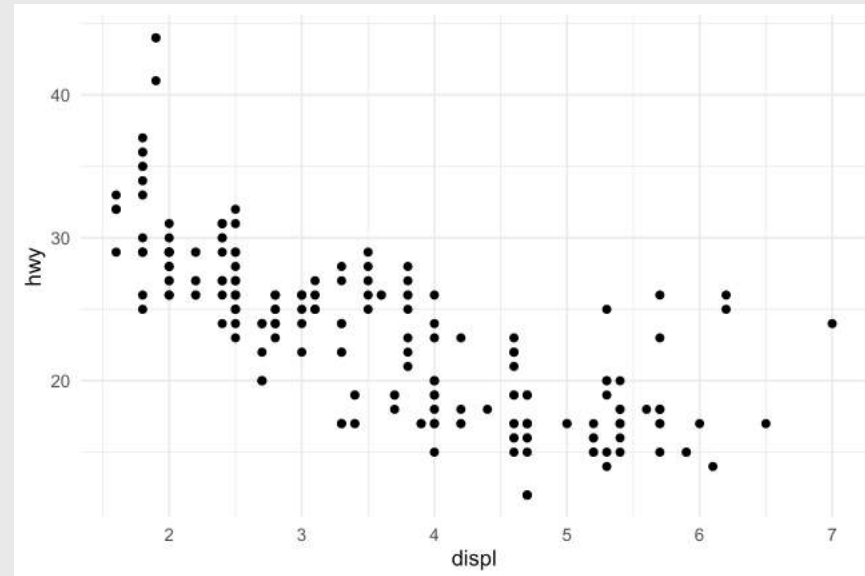
## theme\_bw()

```
mpg %>%  
  ggplot(aes(x = displ, y = hwy)) +  
  geom_point() +  
  theme_bw()
```



## theme\_minimal()

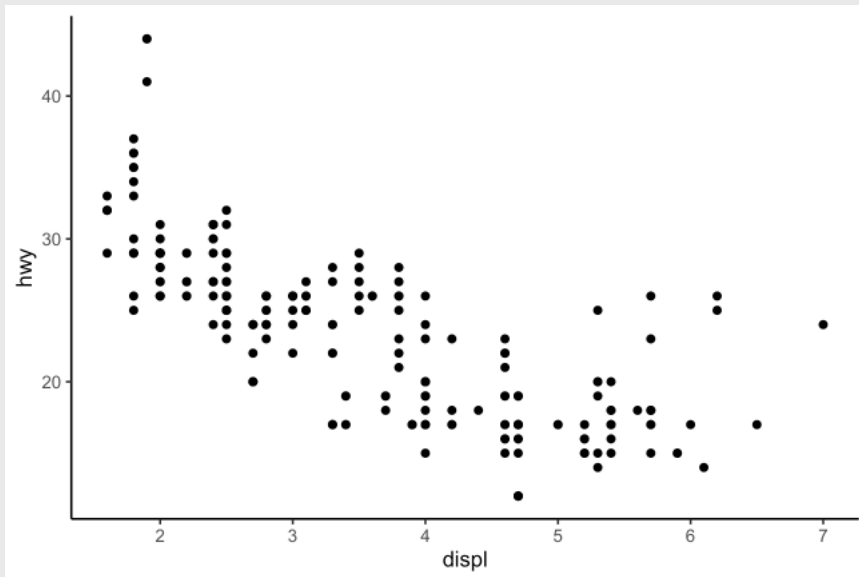
```
mpg %>%  
  ggplot(aes(x = displ, y = hwy)) +  
  geom_point() +  
  theme_minimal()
```



# Common themes

## theme\_classic()

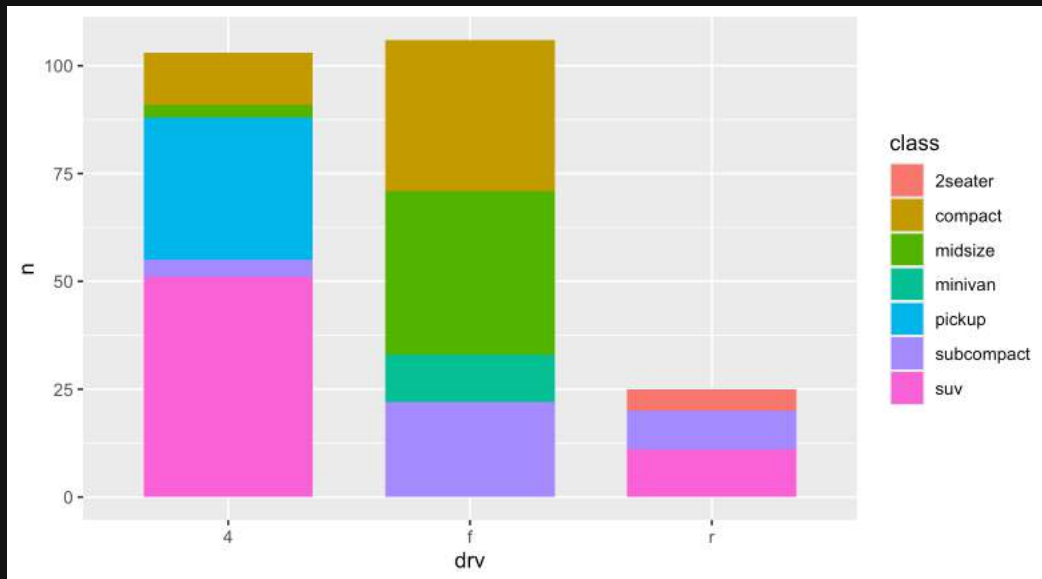
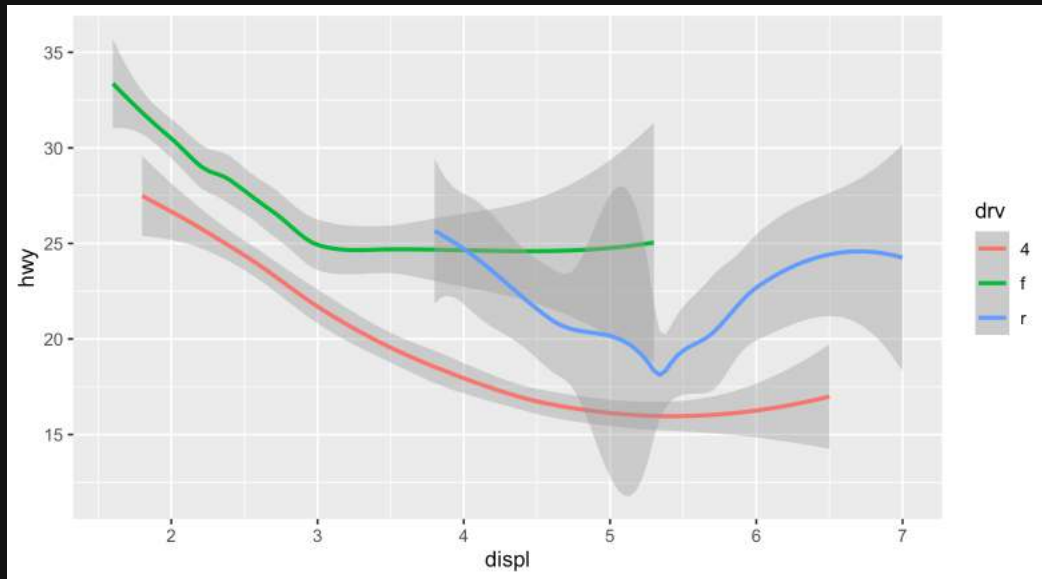
```
mpg %>%  
  ggplot(aes(x = displ, y = hwy)) +  
  geom_point() +  
  theme_classic()
```



## theme\_void()

```
mpg %>%  
  ggplot(aes(x = displ, y = hwy)) +  
  geom_point() +  
  theme_void()
```



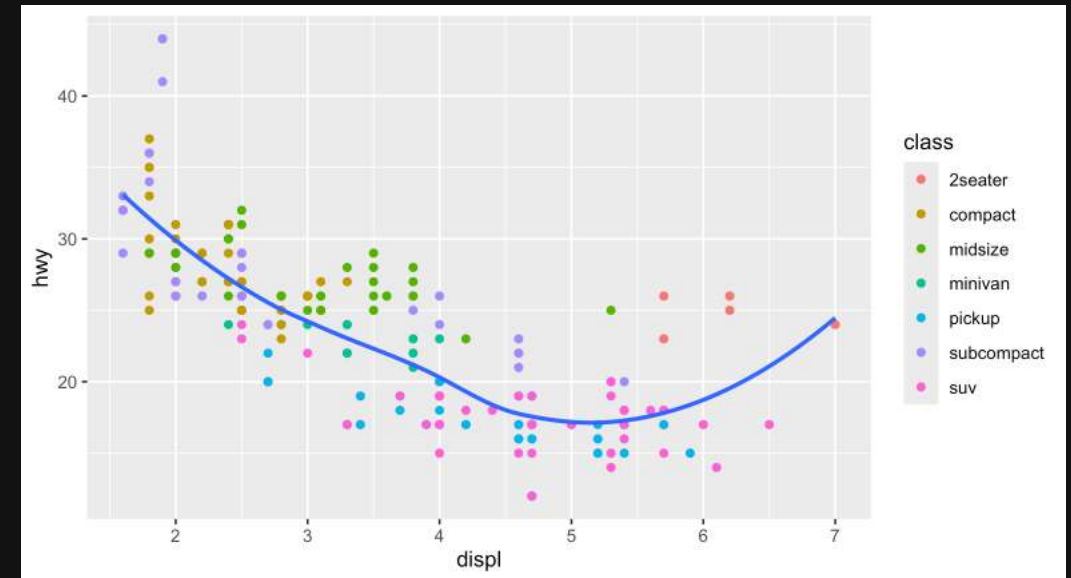


# Your turn

15:00

Open `practice.qmd`

Use the `mpg` data frame and ggplot to create these charts



# Extra practice

